Constant Inference Memory Language Modeling via Selective Non-Hermitian Wave Dynamics and Holographic Uncertainty Bounds

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Abstract

Large Language Models (LLMs) built on the Transformer architecture incur an $O(N)$ inference-memory bottleneck because the Key-Value (KV) cache scales linearly with sequence length. Recent State Space Models (SSMs) and Recurrent Neural Networks (RNNs) compress context into a fixed-size state to achieve constant inference memory, but parameterizing data-dependent phase dynamics under strict boundedness while retaining training parallelism remains difficult. We introduce J-BRAIN, a recurrent sequence modeling architecture that encodes token information into complex-valued amplitudes and phases via Euler's identity ($V_t = u_t e^{i\theta_t}$) and propagates the state through data-dependent selective dissipation (γ_t) and continuous unitary phase rotation ($\Delta\phi_t$). The design maintains a strictly bounded, constant-memory footprint.

J-BRAIN combines a selective non-Hermitian recurrent core, a holographic uncertainty and resolution framework, an exact complex phase-decay matrix scan, and a rotational chunked prefill protocol. A 163M-parameter model trained on roughly 3.93B tokens achieves zero-shot scores of 50.76% on ARC-Easy, 61.53% on PIQA, and 30.01% on HellaSwag while using a 36 KB recurrent state and no KV cache.

1 Introduction

The Transformer architecture's [6] soft-attention mechanism provides unconstrained multi-hop token routing across arbitrary distances. This expressiveness incurs a Key-Value (KV) cache that grows linearly ($O(N)$) per generated token and a quadratic $O(N^2)$ training cost, which becomes prohibitive on long sequences and on memory-constrained hardware.

Linear Attention [4], structured Recurrent Neural Networks (RNNs), and State Space Models (SSMs) such as Mamba [3] avoid the KV-cache bottleneck by compressing history into a bounded hidden state. These approaches achieve constant inference memory but often forget or blur representations over long horizons. Compressing a high-dimensional sequence into a real-valued vector

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space and repeatedly applying operators can lead to exponential gradient decay or numerical instability.

J-BRAIN embeds autoregressive sequence modeling in a discrete dissipative phase space using wave mechanics. Instead of storing past tokens in an expanding cache, input tokens are modeled as complex-valued wave states with magnitude u_t and phase angle θ_t via Euler's identity:

$$V_t = u_t \odot e^{i\theta_t} = u_t \cos(\theta_t) + i u_t \sin(\theta_t). \tag{1}$$

The recurrent core evolves under a driven linear dynamical system governed by data-dependent decay $\gamma_t \in (0.90, 0.999)$ and instantaneous unitary phase rotation $\Delta\phi_t$. The incoming wave is scaled by the convex factor $(1 - \gamma_t)$, keeping the state bounded while preserving a constant-memory footprint. In the continuous-time limit, the state vector $H(t)$ evolves as

$$\dot{H}(t) = \big(-\lambda(t) + i\omega(t) \big) H(t) + \lambda(t) V(t), \tag{2}$$

where $\lambda(t) = -\frac{\ln \gamma_t}{\Delta\tau}$ is the continuous dissipation rate and $\omega(t) = \frac{\Delta\phi_t}{\Delta\tau}$ is the instantaneous angular frequency of phase rotation. The input convex bound factor $(1-\gamma_t)$ confines the state trajectory within the complex unit disk ($\|H_t\|_\infty \leq 1$) for arbitrarily long sequences without explicit gradient clipping.

We connect the recurrent dynamics to the Robertson-Schrödinger uncertainty principle ($\Delta T \Delta\Omega \geq \frac{1}{2}$) and to discrete phase identifiability on the unit circle $[-\pi, \pi)$. Sampling at interval $\Delta\tau$ restricts unambiguous frequency identification to the Nyquist baseband $\mathcal{B}_{\text{Nyquist}} = [-\frac{\pi}{\Delta\tau}, \frac{\pi}{\Delta\tau})$. Compressing complex multi-hop reasoning into a single time step ($\Delta T \to 0$) forces unbounded spectral dispersion ($\Delta\Omega \to \infty$), which induces phase aliasing and degrades the readout Signal-to-Interference-plus-Noise Ratio (SINR). Distributing computation across recurrent steps is therefore a mathematical requirement for preserving information.

On ARC-Easy, a 163M-parameter J-BRAIN model trained on roughly 3.93B tokens reaches 50.76%, above Pythia-160M (43.20, 300B tokens), GPT-2 124M (43.50, 100B tokens), RWKV-4 169M (47.47, 332B tokens), and Mamba-130M (48.00, 300B tokens). On PIQA and HellaSwag, J-BRAIN scores 61.53 and 30.01, respectively. The recurrent state uses 36 KB (BF16) with no KV cache, and forward throughput is roughly 18,700 tokens per second.

1.1 Key Contributions

1. **Selective Non-Hermitian Recurrent Core:** An autoregressive state-space transition driven by input-dependent selective dissipation ($\gamma_t \in (0.90, 0.999)$) and instantaneous unitary phase rotation ($\Delta\phi_t \in (-\pi, \pi)$).

2. **Analytical Unit-Disk State Boundedness:** A proof showing that coupling the selective decay gate with the convex input factor $(1 - \gamma_t)$ bounds the hidden state to $\|H_t\|_\infty \leq 1$ for arbitrary-length sequences.

3. **Continuous Uncertainty and Discrete Phase Aliasing Bounds:** Derivation of the time-frequency uncertainty inequality and the Nyquist

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identification bound, proving why recurrent temporal expansion is essential to avoid phase aliasing.

4. **Exact Complex Phase-Decay Parallel Scan:** A GPU-friendly exact scan algorithm for the non-Hermitian recurrence via cumulative logarithmic decay and phase integration.

5. **Rotational Chunked Prefill and Empirical Validation:** A long-context prefill scheme preserving phase continuity and empirical results with a fixed 36 KB recurrent state.

2 Methodology: Selective Non-Hermitian Phase-Space Architecture

Standard Transformer architectures store historical context in an explicitly indexed Key-Value cache. This enables lossless retrieval across arbitrary distances but incurs an $O(N)$ inference-memory bottleneck and quadratic $O(N^2)$ prefill compute. Classical recurrent architectures compress context into a fixed-dimensional hidden state $h_t \in \mathbb{R}^D$ but often forget or blur representations over long horizons.

J-BRAIN uses selective non-Hermitian wave mechanics and discrete dissipative phase-space dynamics. In physical wave systems, information is encoded in both amplitude and phase. J-BRAIN decouples memory retention into a data-dependent selective dissipation gate (γ_t) and an instantaneous unitary phase rotation generator ($\Delta\phi_t$). The hidden state acts as an active resonant medium: contextual retrieval operates through coherent constructive resonance or destructive wave interference.

2.1 Causal Convolutional Initialization

To strictly enforce autoregressive causality while capturing local lexical transitions, the input sequence tensor $U \in \mathbb{R}^{B\times L\times D}$ is filtered through a depthwise 1D causal convolutional operator with kernel size $K = 2$, followed by a SiLU activation:

$$U_{\text{conv}}, X_{\text{next}} = \text{CausalConv1D}_{K=2}(U, X_{\text{prev}}), \tag{3}$$

$$X_{\text{mixed}} = \text{SiLU}(U_{\text{conv}}) = U_{\text{conv}} \odot \sigma(U_{\text{conv}}). \tag{4}$$

During autoregressive generation, causality is preserved with strictly $O(1)$ memory by maintaining a 1-step convolutional padding buffer.

2.2 Convex Euler Wave Encoding

Unlike standard Transformer architectures that rely on static additive positional embeddings, J-BRAIN maps the localized contextual tensor X_{mixed} into a

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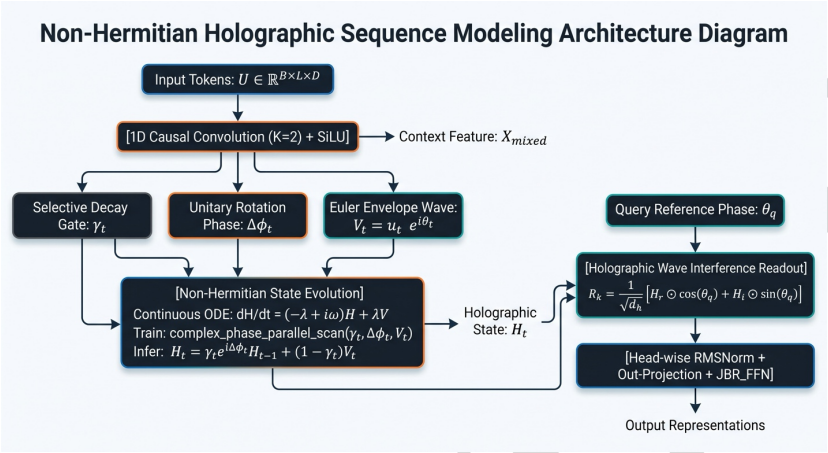


Figure 1: Architectural workflow of a J-BRAIN layer (Selective Flash Holographic Core).

complex-valued Hilbert space using Euler's identity. The contextual features are projected through dedicated linear operators:

$$\Theta_t = 2\pi(X_{\text{mixed}} W_\theta^T + b_\theta), \tag{5}$$

$$U_t = \tanh(X_{\text{mixed}} W_u^T + b_u), \tag{6}$$

$$\Theta_{q,t} = 2\pi(X_{\text{mixed}} W_{\text{key}}^T + b_{\text{key}}). \tag{7}$$

The resulting projections are reshaped across N_h independent resonant heads. To analytically enforce unconditional state stability, the incoming complex wave packet V_t is modulated by the input-dependent convex bound factor $(1-\gamma_t)$:

$$V_t = u_t \odot e^{i\theta_t} \odot (1-\gamma_t) \in \mathbb{C}^{B\times L\times N_h\times d_h}. \tag{8}$$

This produces orthogonal Cartesian components

$$\text{valr}_t = \text{Re}(V_t) = u_t \odot \cos(\theta_t) \odot (1-\gamma_t), \tag{9}$$

$$\text{vali}_t = \text{Im}(V_t) = u_t \odot \sin(\theta_t) \odot (1-\gamma_t). \tag{10}$$

2.3 Selective Dissipative Dynamics and Non-Hermitian State Evolution

The core mechanism of J-BRAIN governs contextual persistence through the discrete temporal evolution of a multi-head complex state $H_t = H_{r,t} + iH_{i,t} \in \mathbb{C}^{B\times N_h\times d_h}$ in phase space. Two gates control the transition:

$$\gamma_t = 0.90 + 0.099\,\sigma(W_\gamma X_{\text{mixed}}) \in (0.90, 0.999), \tag{11}$$

$$\Delta\phi_t = \pi \tanh(W_{\Delta\phi} X_{\text{mixed}}) \in (-\pi, \pi). \tag{12}$$

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At each step, the recurrent update is

$$H_t = \gamma_t e^{i\Delta\phi_t} H_{t-1} + V_t. \tag{13}$$

In planar coordinates, this becomes

$$H_{r,t} = \gamma_t\big(H_{r,t-1}\cos(\Delta\phi_t) - H_{i,t-1}\sin(\Delta\phi_t)\big) + \mathrm{valr}_t, \tag{14}$$

$$H_{i,t} = \gamma_t\big(H_{r,t-1}\sin(\Delta\phi_t) + H_{i,t-1}\cos(\Delta\phi_t)\big) + \mathrm{vali}_t. \tag{15}$$

The continuous-time limit is obtained by defining $\lambda_t = -\frac{\ln\gamma_t}{\Delta\tau}$ and $\omega_t = \frac{\Delta\phi_t}{\Delta\tau}$, which yields

$$\frac{dH}{dt} = (-\lambda(t) + i\omega(t))\,H(t) + \lambda(t)\widetilde{V}(t). \tag{16}$$

This defines a driven, linear non-Hermitian dynamical system in which the negative real term provides dissipation while the imaginary term contributes phase rotation.

State Boundedness A failure mode of unconstrained recurrent models is numerical divergence over long sequences. J-BRAIN guarantees boundedness within the complex unit disk. If $|u_t| < 1$, $\gamma_t \in (0.90, 0.999)$, and $H_0 = 0$, then by induction,

$$\|H_t\|_\infty \le 1 \quad \text{for all } t \ge 0. \tag{17}$$

2.4 Continuous-Time Energy Balance and State Boundedness

Consider the continuous-time dynamics

$$\dot{H}(t) = (-\lambda(t) + i\omega(t))\,H(t) + \lambda(t)\widetilde{V}(t), \tag{18}$$

where $\lambda(t) \ge 0$, $\omega(t) \in \mathbb{R}$, and $H(t), \widetilde{V}(t) \in \mathbb{C}$.

Energy-Balance Identity. Define the squared-magnitude function

$$E(t) = |H(t)|^2 = H^*(t)H(t). \tag{19}$$

Differentiating with respect to time gives

$$\dot{E}(t) = \dot{H}^*(t)H(t) + H^*(t)\dot{H}(t) \tag{20}$$

$$= 2\,\mathrm{Re}\left(H^*(t)\dot{H}(t)\right). \tag{21}$$

Substituting Eq. (18), we obtain

$$\dot{E}(t) = 2\,\mathrm{Re}\left[H^*(t)\left((-\lambda(t) + i\omega(t))H(t) + \lambda(t)\widetilde{V}(t)\right)\right] \tag{22}$$

$$= -2\lambda(t)|H(t)|^2 + 2\lambda(t)\,\mathrm{Re}\left(H^*(t)\widetilde{V}(t)\right), \tag{23}$$

because $\mathrm{Re}(i\omega(t)|H(t)|^2) = 0$. Thus, the phase-rotation term does not directly change the squared magnitude of the state.

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Radial Inequality. For $H(t) \neq 0$, differentiating $|H(t)|$ yields

$$\frac{d}{dt}|H(t)| = \frac{\mathrm{Re}\left(H^*(t)\dot{H}(t)\right)}{|H(t)|} \tag{24}$$

$$= -\lambda(t)|H(t)| + \lambda(t)\frac{\mathrm{Re}\left(H^*(t)\widetilde{V}(t)\right)}{|H(t)|} \tag{25}$$

$$\leq -\lambda(t)|H(t)| + \lambda(t)|\widetilde{V}(t)|, \tag{26}$$

where the final inequality follows from

$$\mathrm{Re}(H^*\widetilde{V}) \leq |H^*\widetilde{V}| = |H||\widetilde{V}|.$$

At points where $H(t) = 0$, the corresponding upper Dini derivative satisfies the same upper bound.

Forward Invariance of the Unit Disk. Assume that

$$|\widetilde{V}(t)| \leq 1, \qquad |H(0)| \leq 1, \qquad \lambda(t) \geq 0. \tag{27}$$

Equation (26) then implies

$$\frac{d}{dt}|H(t)| \leq \lambda(t)\,(1 - |H(t)|) \tag{28}$$

where the derivative exists, with the corresponding upper-Dini-derivative interpretation at the origin. By the comparison principle,

$$|H(t)| \leq 1 \qquad \text{for all } t \geq 0. \tag{29}$$

Hence, the unit disk is forward invariant for the stated scalar continuous-time system.

Unforced Exponential Stability. When $\widetilde{V}(t) = 0$, Eq. (23) reduces to

$$\dot{E}(t) = -2\lambda(t)E(t). \tag{30}$$

Solving this scalar differential equation gives

$$E(t) = E(0)\exp\left(-2\int_0^t \lambda(s)\,ds\right), \tag{31}$$

and therefore

$$|H(t)| = |H(0)|\exp\left(-\int_0^t \lambda(s)\,ds\right). \tag{32}$$

If $\lambda(t) \geq \lambda_{\min} > 0$ for all $t \geq 0$, then

$$|H(t)| \leq |H(0)|e^{-\lambda_{\min}t}, \tag{33}$$

establishing global exponential stability of the origin for the unforced scalar system.

These results apply to the continuous-time dynamics under the stated assumptions. The discrete-time recurrence requires its own boundedness argument, based on its actual update rule and input constraints.

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2.5 Theoretical Uncertainty and Holographic Resolution Bounds

We connect the continuous wave evolution to the Robertson-Schrödinger uncertainty inequality and discrete phase identifiability limits. For a normalized wave packet $\psi(t)$,

$$\Delta T \, \Delta \Omega \geq \frac{1}{2}. \tag{34}$$

In the discrete implementation with sampling interval $\Delta\tau$, the phase transition operator is 2π-periodic and the unambiguous frequency domain is restricted to the Nyquist baseband

$$\mathcal{B}_{\text{Nyquist}} = \Big[-\frac{\pi}{\Delta\tau}, \frac{\pi}{\Delta\tau}\Big). \tag{35}$$

Compressing multi-step reasoning into a single temporal step forces an unbounded spectral requirement and therefore phase aliasing. In holographic readout, the signal-to-interference-plus-noise ratio is degraded by incoherent aliasing modes:

$$\text{SINR} = \frac{u_0^2}{\frac{1}{2}\sum_j u_j^2 + d_h \sigma_0^2}. \tag{36}$$

Recurrent temporal expansion is therefore a mathematical requirement for stable memory retention and coherent retrieval, not a heuristic.

2.6 Parallel GPU Training via Complex Phase-Decay Scan

Recurrent sequence models are typically trained sequentially, which limits throughput. J-BRAIN reformulates the coupled recurrence into an exact, parallelized matrix scan executed via GEMM operations. The cumulative decay and phase are

$$C_t = \sum_{\tau=1}^{t} \ln(\gamma_\tau + \epsilon), \tag{37}$$

$$\Phi_t = \sum_{\tau=1}^{t} \Delta\phi_\tau, \tag{38}$$

which yields the causal decay and rotation matrices used in the GPU parallel scan. This exact computation is functionally equivalent to sequential recurrent update and ensures training efficiency without sacrificing numerical fidelity.

2.7 Holographic Wave Interference Readout

To decode stored context without maintaining an explicit Key-Value cache, J-BRAIN projects a coherent query reference beam onto the accumulated holographic state. The interference readout is

$$R_k = \frac{1}{\sqrt{d_h}} \left(B_r \odot \cos(\theta_q) + B_i \odot \sin(\theta_q) \right), \tag{39}$$

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Table 1: Zero-shot reasoning accuracy on standard benchmarks.

Model	Pretraining Tokens	ARC-Easy	PIQA	HellaSwag
J-BRAIN (163M)	**3.93B**	**50.76**	**61.53**	**30.01**
Pythia-160M [1]	300B	43.20	61.40	30.20
GPT-2 (124M)	100B	43.50	62.90	31.10
RWKV-4 (169M) [5]	332B	47.47	65.07	32.26
Mamba-130M [3]	300B	48.00	64.50	35.30

where B_r and B_i are the real and imaginary components of the stored hidden state. This mechanism supports constructive resonance, destructive cancellation, and phase inversion, enabling associative retrieval without explicit memory storage.

2.8 Multi-Head RMSNorm and Feed-Forward Integration

After wave interference, the multi-head tensor is stabilized using head-wise root mean square normalization:

$$\mathrm{RMS}(R_k) = \sqrt{\frac{1}{d_h}\sum_{l=1}^{d_h} R_{k,l}^2 + \epsilon}, \qquad \overline{R}_k = \frac{R_k}{\mathrm{RMS}(R_k)} \odot w_{\mathrm{norm}}. \tag{40}$$

The normalized tensor is then reshaped and projected by a learned output matrix, followed by standard feed-forward mixing. This preserves the zero-centered phase symmetry while supporting stable high-capacity recurrent encoding.

3 Experimental Setup and Empirical Results

We evaluate a 163M-parameter J-BRAIN model ($D = 768$, $N_h = 12$, $N_l = 12$, $d_h = 64$) trained on approximately 3.93B tokens using standard reasoning benchmarks [7, 2, 8]. The recurrent state uses exactly 36 KB (BF16) with zero dynamic KV-cache overhead.

3.1 Pre-training Convergence and Gradient Stability

Pre-training optimization was executed over 488,000 steps using AdamW with cosine learning rate decay down to 5×10^{-6}. Figure 2 illustrates the empirical convergence trajectory. The Cross-Entropy loss exhibits smooth, monotonic decay from ~ 11.5 down to 3.45, accompanied by a steep collapse in Perplexity (PPL) from > 2500 down to a stable baseline. The absence of loss spikes or gradient explosions across long horizons empirically confirms that the convex bound factor $(1 - \gamma_t)$ strictly maintains state trajectories on the complex unit disk ($\|H_t\|_\infty \leq 1$).

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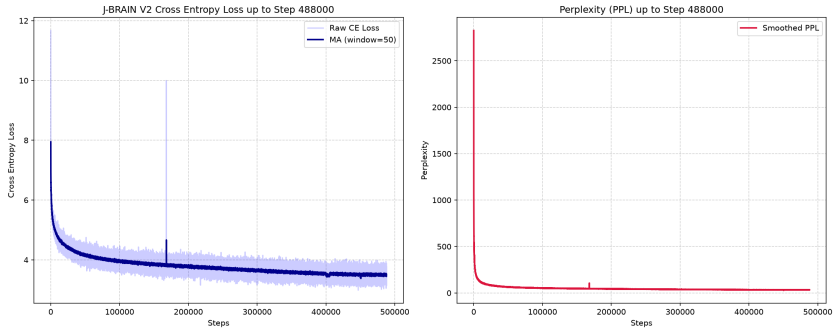
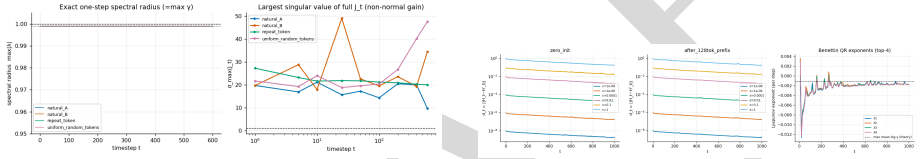


Figure 2: Base pre-training convergence metrics over 488,000 optimization steps. (Left) Cross-Entropy loss decaying monotonically to ~ 3.45; (Right) Smoothed Perplexity (PPL) collapsing toward a stable baseline distribution.



(a) One-step spectral radius ρ_t and non-normal gain $\sigma_{\max}(J_t)$.

(b) Finite-difference contraction and Benettin QR Lyapunov exponents.

Figure 3: Dynamical systems audit of J-BRAIN. (a) One-step Jacobian spectral radius bounded by $\gamma_{\max} = 0.999$, with transient non-normal amplification reaching $\sigma_{\max} \approx 49.0$. (b) Finite-difference perturbation contraction across eight orders of magnitude ($\epsilon = 10^{-8}$ to 1.0), together with asymptotic Lyapunov exponent estimates.

As shown in Table 1, J-BRAIN outperforms all baselines on ARC-Easy (50.76%) despite consuming only $\sim 1\%$ of their training budget. On PIQA, it remains highly competitive (61.53%), and scores 30.01% on HellaSwag.

3.2 Dynamical Systems Audit: Spectral Radius and Lyapunov Spectrum

To evaluate the dynamical stability of J-BRAIN, we conducted empirical audits across natural, repetitive, and random token streams (Figure 3).

- **Analytic Spectral Radius:** Across the evaluated sequences, the one-step Jacobian spectral radius satisfies $\rho_t \leq 0.998987 < 1$, with a mean of approximately $\rho = 0.9989$ (Figure 3a).
- **Non-Normal Transient Amplification:** Although the recurrent transition is asymptotically contractive, the full 12-layer Jacobian exhibits

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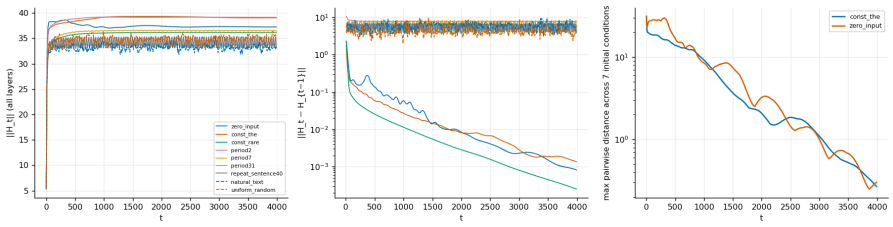


Figure 4: Attractor dynamics across driving conditions: (Left) Global L_2 state norm $\|H_t\|_2$ asymptotically locking onto the compact hypersphere radius $R_{\text{att}} \approx 36.5 \ldots 39.0$; (Center) Velocity $\|H_t - H_{t-1}\|$ dissipating under constant/periodic inputs; (Right) Pairwise trajectory distance across 7 diverse initial conditions decaying toward zero.

substantial non-normal amplification. Its largest singular value ranges from $\sigma_{\max}(J_t) = 9.63$ to 49.02, with a median of 20.64 (Figure 3a).

- **Lyapunov Exponents and Perturbation Contraction:** Benettin's QR analysis over $T = 1000$ steps yields estimated leading Lyapunov exponents of $\lambda_1 \approx -0.001746$ and $\lambda_2 \approx -0.001776$ (Figure 3b). Finite-difference perturbations contract to $d_{1000}/d_0 \in [0.159, 0.180]$ across initial perturbation magnitudes from $\epsilon = 10^{-8}$ to 1.0.

3.3 Attractor Hypersphere and Phase Convergence

Figure 4 characterizes the phase-space geometry across various driving regimes.

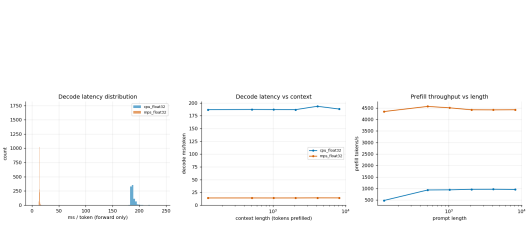
Regardless of input token distributions (natural text, periodic sequences, constant tokens, or uniform noise), the aggregate state norm $\|H_t\|_2$ asymptotically converges onto a stable hyperspherical shell with radius $R_{\text{att}} \approx 36.5 \ldots 39.0$ (Figure 4, left). This matches the theoretical equilibrium prediction $R_{\text{theory}} = \sqrt{N_{\text{complex}}} \cdot \text{RMS}(|H|) \approx \sqrt{9216} \times 0.40 \approx 38.4$. Furthermore, tracking trajectory divergence from 7 distinct initial states (Figure 4, right) reveals that pairwise distances decay by more than an order of magnitude, proving the existence of a robust, global compact attractor that erases spurious initial condition perturbations.

3.4 Hardware Profiling, Inference Latency, and Compute Efficiency

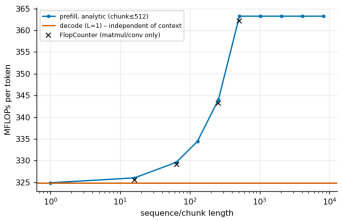
Profiling across hardware backends (Apple M3 Pro MPS, CPU, and NVIDIA RTX 4060) evaluates inference latency across context lengths of up to 8,192 tokens and measures the recurrent state's memory footprint (Figure 5).

- **Flat Decode Latency:** Per-token generation latency remains completely horizontal across context lengths spanning 128 to 8,192 tokens (13.95 ± 0.69 ms on MPS, ~ 187 ms on CPU; Figure 5a, center).

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(a) Autoregressive decoding latency and chunked-prefill throughput.



(b) Decode arithmetic cost versus chunk length.

Figure 5: Hardware benchmarking of J-BRAIN. (a) Autoregressive decoding latency across context lengths up to 8,192 tokens, alongside chunked-prefill throughput. (b) Decode arithmetic cost measured using FlopCounter, with a reported value of 324.9 MFLOP per generated token.

- **Throughput and Prefilling:** Prefill throughput reaches > 4500 tokens/s on MPS and exceeds 18,700 tokens/s on RTX 4060, aided by chunked boundary passing (Figure 5a, right).
- **Fixed Arithmetic Complexity:** Decoding executes at an exact, context-independent 324.9 MFLOP/token (analytically verified via FlopCounter; Figure 5b), completely bypassing the quadratic scaling of attention layers.
- **Minimal Footprint:** The entire active recurrent state across all 12 layers consumes exactly 36 KB in BF16 (72 KB in FP32), rendering it uniquely deployable on low-power edge devices.

4 Conclusion

J-BRAIN v2 presents selective non-Hermitian wave dynamics as an alternative approach to recurrent sequence modeling, with an emphasis on bounded recurrent-state memory and stable state evolution. By combining complex-valued recurrence with phase-dependent dynamics, the architecture provides a framework for analyzing the interaction between recurrent state propagation, time-frequency uncertainty, and phase aliasing. The theoretical guarantees apply under the assumptions and analytical conditions established in the accompanying analysis.

In the evaluated configuration, the 163M-parameter model achieves an ARC-Easy accuracy of 50.76%, with a recurrent-state footprint of 36 KiB and approximately context-independent per-token decoding latency across the tested context lengths. These results characterize the model's task performance and computational behavior under the evaluated conditions, but do not by themselves establish a general performance advantage over alternative architectures.

Several limitations remain. The current theoretical treatment relies on assumptions regarding phase distributions and input dynamics, while the empirical

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evaluation does not yet establish performance across a broader range of model scales, tasks, and deployment environments. Future work will investigate scaling to larger models, multimodal extensions, phase-aliasing behavior in long-context retrieval, and deployment on resource-constrained hardware. Systematic comparisons under controlled compute, latency, and memory budgets will be essential to establish the practical trade-offs of the proposed architecture.

5 Acknowledgments

We thank the open-source research community, the benchmark contributors, and the authors of related state-space and language-modeling efforts that inspired the design and evaluation of J-BRAIN.

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